

CHAPTER 2

History

2.1. Space story

My father took my brother and me to a showing of Stanley Kubrick's movie, "2001: A Space Odyssey" at the Cinerama Theater in New York City. The movie was filmed in Super Panavision 70 with some scenes in Todd-AO and MCS-70. The Cinerama theater showed the film on a curved screen helping to immerse the audience in the film. The theater itself was one of the first "movie palaces" that looked like a theater where a ballet or opera might be performed but were designed only for movies. The first 20 minutes were a surprise as were the last 20 minutes or so. In between, the movie was an excellent tutorial on the future of spaceflight.

2.2. Introduction

It is hard to be a spacecraft engineer unless you know the history of spacecraft engineering. Knowing history allows you to learn from past successes and mistakes. It also helps prevent you from "reinventing the wheel" instead of taking what has been done and building on that with your innovations. It is also a lot of fun to see what engineers and scientists did when they were first learning the elements of spacecraft design. Many of the engineers went from designing aircraft to spacecraft. While spacecraft and aircraft are different, the elements of design and the methodologies of design do not change. The cancelation of the Canadian supersonic C-105 Arrow fighter program resulted in several very talented Canadian engineers moving to the United States to work on Apollo. This chapter will include comments on the control aspects of the designs.

We will start with ancient history. Some will be in the realm of fiction with the first ideas of how one might travel in space and what adventures might be found. The remaining sections will cover the "Space Race" and the beginnings of automated spacecraft and robotic exploration. Engineers were learning how to make spacecraft autonomous back in the 1950s and 1960s without the use of digital computers. Digital computers enabled the Apollo missions. The chapter will continue to the present day and give some ideas on where the space business will go in the future.

2.3. Pre-1950 – dreaming of space

Robert Goddard pioneered rockets, closely followed by the Germans led by Werner von Braun. The first applications were military, the V-2 for example. However, all of

the pioneers were interested in space. Sängers' Antipodal bomber, a boost-glide vehicle designed to bomb the United States, was also a potential launch vehicle. It was the inspiration for the German Säger II launch vehicle. Säger II was to be fully reusable with a hypersonic turboramjet-powered first stage and an LH_2/LO_2 second stage. It would have flown from Europe to the equator at Mach 4 and then accelerated to Mach 6.7 before releasing the second stage. It was canceled because it did not appear to offer any cost advantages over the Ariane 5.

A multistage V-2 might have reached orbit, and later effectively did as part of both the US and Soviet space programs. The first successful launch vehicles had analog control systems and very few ways to test them, aside from actually flying the vehicle. The number of failures should be thought of in terms of how many bugs there are in the average modern software package.

2.4. 1950s – getting started

Intercontinental Ballistic Missiles (ICBMs) drove the space programs of both the United States and the Soviets. While bombers could deliver nuclear weapons, they were vulnerable to ground-to-air missiles and fighters. ICBMs gave an opponent only minutes to make a decision and shooting one down remains an elusive goal to this day. Soviet bombs were bigger than US bombs so they built bigger boosters, which gave them an edge in the early days of the space race. Sputnik was launched, alarming the United States and leading to the Space Race.

2.5. 1960s – the Golden Age: Apollo to the moon

Vostok 1 kicked off the space race big time, even more than Sputnik. Space was in the news and on TV in the 1960s. It was so important that engineers and scientists involved in the programs were celebrities. Hence the term “Rocket Scientist”.

Both the United States and Soviet Union launched crewed and robotic missions. Those with crews included the Vostok, Soyuz, Mercury, Gemini, and Apollo spacecraft. The robotic missions included the first landings on the moon and missions to Venus and Mars. Oddly, Venus has been a very successful Russian destination, while Mars has been pretty much a US destination.

Commercial activities began with the first commercial communications satellites, which remain a big and profitable business. It may still be the only profitable space business. Three American companies, Ford Aerospace, RCA Astro-Electronics, and Hughes Space and Communications were the leaders in the communication-satellite business in the 1960s but the Europeans and Japanese followed quickly with their communication satellites. Now, communication-satellite operators have a huge number of choices for their communication satellites.

The US Apollo moon program was a driving force in space in the 1960s. One of the first issues in human spaceflight was how much control, if any, to give the pilot. The engineers were quite confident that their automatic control systems could work successfully without human intervention. They were concerned the pilot would make any situation worse. This belief was a fundamental error based on a misunderstanding of control. You only need closed control to deal with uncertainties in the system. Pilots are adaptable. Their control systems were not. At least in the US program, the crews were essential.

Apollo had the first fully digital control system. The Charles Stark Draper Laboratory had to build its computer from medium-scale integrated circuits such as a NAND gate on a chip. Memory was magnetic cores woven by textile workers. Draper also designed the programming languages. The flight computer could not be reprogrammed in flight, leading to numerous workarounds. One of the most famous was the stuck abort switch on Apollo 14. In a very short amount of time, MIT software engineers came up with a workaround to convince the flight computer that the landing had already been aborted, thus preventing from aborting the landing, which went on as scheduled. Don Eyles, the lead on the team that solved the problem, was awarded a medal by the City of Boston for his quick and innovative work. Both the Apollo Command Module and Lunar Module used phase-plane controllers. These were easy to implement using digital computers. Example 2.1 shows a phase-plane controller that is similar to the Lunar Module system. The Lunar Module had a clever system, developed by Richard Goss [1], that allowed a 3-axis phase-plane controller to work effectively given the Lunar Module thruster geometry.

An important aspect was manual control of the Lunar Module. Robert Stengel [2] led its design. Humans in the loop are equally as important on remote spacecraft, though typically a lot less thought is given to the human/spacecraft interface for robotic spacecraft.

It is easy to forget that before Apollo, controls engineers thought in terms of block diagrams that could be turned into analog control systems using operational amplifiers. This led to a long list of block diagram-oriented products like Control-C, Matrix-X, and Simulink. Unfortunately, much like you can write spaghetti code, you can write spaghetti block diagrams. As flight-control systems start using expert systems and neural networks, block diagrams will be even less useful in the future.

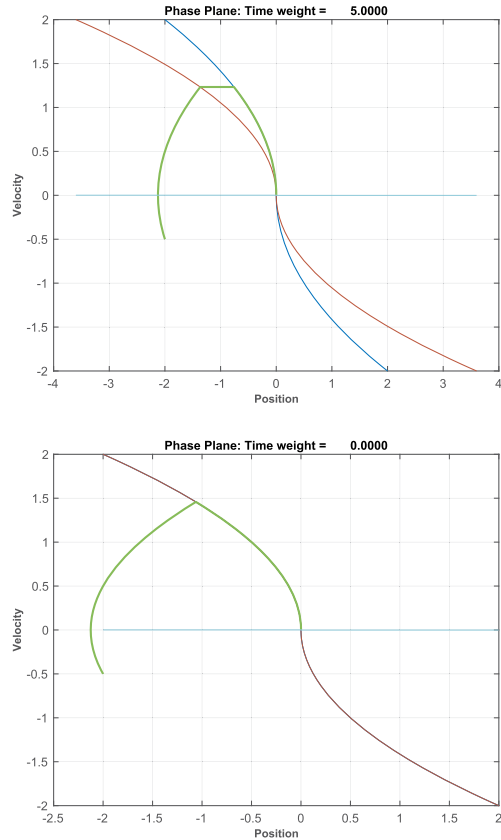
NASA had ambitious plans for the moon including a nuclear thermal upper stage for the Saturn V. Early versions of nuclear thermal stages were ground tested. Robotic versions of the Lunar Module were planned for landing large masses of cargo. Unfortunately, the Apollo program was canceled after Apollo 17 due to the perception that it had lost public support.

The Defense Meteorological Satellite Program (DMSP) began in the 1960s at RCA Astro-Electronics in NJ. The satellites went through many changes over their long

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1 n           = 8000;
2 xP          = zeros(3,n);
3 dT          = 0.001;
4 x           = [-2;-0.5];
5 dRHS        = struct('inr',3,'u',0);
6 kOpt        = 0;
7
8 for k = 1:n
9     u = PhasePlane(x(2),x(1),kOpt,0.001,0.001);
10    xP(:,k) = [x;u];
11    dRHS.u = u;
12    x = RK4(@RHS,x,dT,0,dRHS);
13 end
14
15 PhasePlanePlot( xP(2,:), xP(1,:),kOpt
16               ,0.001,0.001,[-2,2])
17
18 %% Right hand side [ang1;ang2;omega1;omega2]
19 function xDot = RHS(x,~,d)
20
21 xDot      = [x(2);d.u];
22
23 end

```



Example 2.1: An Apollo Lunar Module-type phase-plane controller. The red and blue lines are the optimal switching lines. They are separated in the first plot because of the fuel consideration of the optimal control. The second plot shows the time optimal switching curves.

history. The TIROS satellite was the NASA version. Eventually, they merged into a common spacecraft. The NASA version used an Earth sensor, while the Air Force system had a star camera and an IMU-based Kalman filter for attitude determination. The star camera was a v-slit design and identified stars using the star transits of the slits. It had an 80-star catalog.

2.6. 1970s – the Space Shuttle era

Skylab and the Apollo-Soyuz missions were the immediate follow-ons to Apollo using leftover hardware. The Space Shuttle was the follow-on to Apollo. NASA had its eyes on lunar bases and human missions to Mars but ended up with the Shuttle that was sup-

posed to lower the cost of getting payloads and humans into space, thus enabling lunar and Mars missions. The Shuttle, despite two tragic failures, was a successful program. The Shuttle launched Hubble and many modules for the International Space Station (ISS). Another program, GPS, was launched in 1973. Arguably it has had the greatest impact of any space program aside from communications satellites. The first US space station, Skylab, was launched and had three different crews. The Soviets launched their Molniya space station. Many Shuttle-centric programs, including a reusable upper stage, were started and then canceled after the Challenger explosion. The Air Force, which was the reason for the size of the Shuttle and its wing shape, bailed out of the Shuttle program as they never really wanted to use it under any circumstances.

The Shuttle was developed at the beginning of the computer era. Simulations for the Shuttle Orbiter ran on mainframes and were limited to about 90 seconds of simulated time. The Shuttle Landing simulator at Draper Laboratory was an analog computer simulation. One of the Draper software engineers said, “The only thing you can be sure about with an analog computer is that today’s results won’t match yesterday’s results.”

The Shuttle introduced the HAL/S (High-order Assembly Language/Shuttle) programming language [3]. The official story is that it is named after Hal Laning of Draper Lab but by coincidence, HAL/9000 of 2001 fame was a notable Artificial-Intelligence movie character of the time. It was a high-level language with innovations such as explicit type definition, long variable names, three lines of print per line of code (for exponents and subscripts), and other interesting features.

```

E      2      2
M  X = A  +  B
S                               |

```

The equivalent MATLAB[®] code would be

```
x = a^2 + b(i)^2;
```

The Space Shuttle also used an IBM 360-based flight computer. It had limited storage and every change of mission phase required a new load of software into the flight computer.

HAL/S was a competitor for the Air Force standard language. Ada (named after Ada Lovelace the first programmer), based on Pascal, ended up winning. HAL/S was used on the NASA Galileo spacecraft [4] but that was the last major application. C, a Bell Labs creation, and later C++ and object-oriented version, eventually won the race for standard languages.

From a control point of view, the Shuttle has considerable heritage from the Apollo GN&C system. It continued to use the phase-plane control system invented for Apollo. The addition of the arm required that the phase-plane update its angular increments (the amount of attitude angle changed per pulse) every time the arm moved since the inertia changes as the arm moves.

The Space Shuttle itself was an international program. The Canadian Space Agency supplied the robot arm, called Canadarm, which was an incredible achievement in robotics. It also taxed the Shuttle's on-orbit autopilot that was not designed with a huge, flexible structure in mind that also changed the Orbiter inertia matrix whenever it moved.

The Voyager spacecraft were launched and are still in operation today. They entered interstellar space and, for the moment, are the first human-built objects that any interstellar space-faring races will encounter, though the NASA New Horizons mission is catching up!

2.7. 1980s – internationalization

The Soviet/American space race was the focus of the 1960s and into the 1970s. Arianespace was founded in 1980 to compete with the US, Soviet, and Chinese launch industries. It was the first major European space program. Since then the European presence in space has grown dramatically. At the same time, Japan's and India's programs grew. Anything the United States could do, they could do. China began working on its program and despite the best efforts for "export control" launched a space station and the first lunar far-side lander. India has been particularly adept at developing low-cost missions. China has also flown its space station.

The 1980s saw competition between RCA, Hughes, and Ford for the communication-satellite market. Hughes had a dual-spin spacecraft, the HS 376, to compete with RCA's momentum-bias Series 2000 through 5000 designs. All three companies supplied most of the world's comsats. All three used spinning parts to give their spacecraft gyroscopic stiffness. Bias momentum is very handy when your spacecraft loses Earth lock because it will not tumble into a random orientation without the Sun on the solar panels.

The Iridium constellation was conceived in 1987 and launched in 1998. It was 66 satellites reduced from the original 77, hence the name. It flew just in time to compete with cell towers so it became a financial failure. It was saved because the US military was using it for communications for forces in remote locations. Its latest version tracks commercial aircraft around the globe along with its communication payload.

2.8. 1990s – Hubble

The Hubble Space Telescope was based on US Air Force telescopes except that it pointed away from the Earth. It was designed to fly on the Space Shuttle mostly because the Space Shuttle was designed to carry similar-sized reconnaissance satellites. Hubble was designed to be maintained by astronauts. This was a good thing because it launched with astigmatism. It was repaired by astronauts several times and continues to operate

to this day. The Hubble used reaction wheels for control. The reaction wheels, which use brushless DC motors, were so noisy that they had to be put on suspensions. Fig. 2.1 shows Hubble. The James Webb Space Telescope uses a similar system.



Figure 2.1 Hubble Space Telescope. Image courtesy of NASA.

Gravity Probe B, a Stanford satellite, was built by Lockheed Martin to test Einstein's General Theory of Relativity. It had a cryogenic Inertial Measurement Unit (IMU) and was so noise sensitive that reaction wheels were ruled out. Instead, it used the helium boil-off from the Inertial Measurement Unit (IMU) dewar for control. The mission was a spectacular success and demonstrated, once again, that General Relativity is correct.

In 1992 Mars Observer was launched. It used elements of the DMSP control system and some features of RCA/GE communications satellites. It was the first real attempt to commercialize space-science missions, that is to have an outside vendor design and build an interplanetary spacecraft with minimal government oversight. Mars Observer was the first interplanetary satellite with reaction wheels. It failed on Mars entry when the propulsion system exploded. The attitude control system needed a last-minute fix to deal with slosh in the fuel tanks. A full-state feedback controller was replaced with a simpler PID controller with roll-off filters to prevent slosh interaction.

1998 was the year of the launch of the central module of the International Space Station (ISS), shown in Fig. 2.3. This would grow into the largest human-built object in space. It also demonstrated even more than Hubble, that humans could work in free fall. The ISS uses double-gimbaled control-moment gyros (CMGs) for control. It also has a reaction control system, which is a satellite that uses thrusters. CMGs require complex

steering logic to keep them out of gimbal lock. Single-gimbal CMGs are often used on Earth-pointing spacecraft in LEO that need to track a target on the ground. Fig. 2.2 shows the ISS CMGs.



Figure 2.2 Four ISS Control-Moment Gyros. Image courtesy of NASA.

You might also observe that the ISS cannot be modeled as a rigid body with all the flexible appendages such as the solar wings and the radiators. Once you add the robots crawling up and down the trusses, the robot arms, etc., it becomes a control engineer's nightmare.

Fig. 2.4 shows the Japanese H-II transfer vehicle. It docks to the ISS via a robot arm.

1999 was the year of the CubeSat Program [5]. CubeSat standard was created by California Polytechnic State University and Stanford University's Space Development Laboratory. Professors Jordi Puig-Suari of Cal Poly and Bob Twiggs of Stanford also realized that small satellites made excellent student projects, and could be useful in their own right. Publishing of a standard simplified the design of CubeSats. The first CubeSats were launched in 2003. This spawned a large and growing industry. One important aspect was that CubeSats greatly lowered the cost of entry to space, leading to innovations in every aspect of space flight.

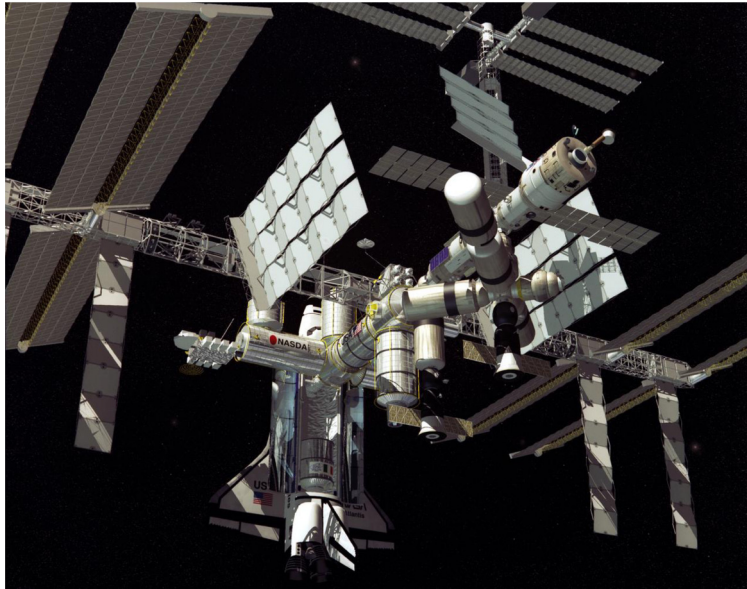


Figure 2.3 The International Space Station. Image courtesy of NASA.

2.9. 2000s – commercial space is reborn

CubeSats started booming with the benefit of helping to train a new generation of space engineers. People started companies for commercial space imaging, commercial science, and other commercial ideas. Space X was founded in 2002. After three launch failures, Falcon 1 successfully reached orbit. This made it feasible for NASA to award a contract to Space X to build the Falcon 9 that has become a space-launch workhorse. While arguably a government-funded entity, Space X works essentially on its own without dozens of NASA design reviews.

The ISS continued to be built, becoming the largest space project ever. Modules from many countries were integrated into the ISS. Japan supplied the JEM module. The ISS was resupplied with the Space Shuttle, Russian Progress spacecraft, the Japanese H-II Kountori Transfer Vehicle, the Dragon resupply vehicle, and the ESA ATV.

The United States continued its exploration of Mars with landers and orbiters. JPL's rovers have covered wide swaths of territory and proven to be reliable explorers. New Horizons, the Pluto explorer, was launched in 2009.

2.10. 2010s – the space station and beyond

The United States continued its exploration of Mars. New Horizons flew by Pluto on July 14, 2015, and continues to explore the edge of the solar system. The very am-



Figure 2.4 Japanese H-II transfer vehicle. The use of space robot arms was pioneered by NASA and the Canadian Space Agency for the Shuttle. Image courtesy of NASA.

bitious James Webb Space Telescope was started and launched in 2021. The United States looked into a variety of follow-ons to the Shuttle. The Orion space capsule and Space Launch System was started. At the same time, the number of commercial entities involved in space grew.

2.11. The future

Five modules are still to be launched to ISS. The United States has funded commercial crewed and robotic lunar landers. There is talk of human Mars missions. China plans to send crews to the Moon and Mars. Interest in interstellar missions, probably robotic, has grown with some NASA money going into preliminary studies. The limitation to more ambitious missions is propulsion, which is difficult. Some money is being spent on nuclear electric and fusion propulsion systems that have the potential to revolutionize space exploration. Fig. 2.5 is a NASA-funded design for a Pluto orbiter powered by a fusion rocket engine. SpaceX wants to send people to Mars by 2029. NASA hopes to land people on the Moon, again, around the same time.

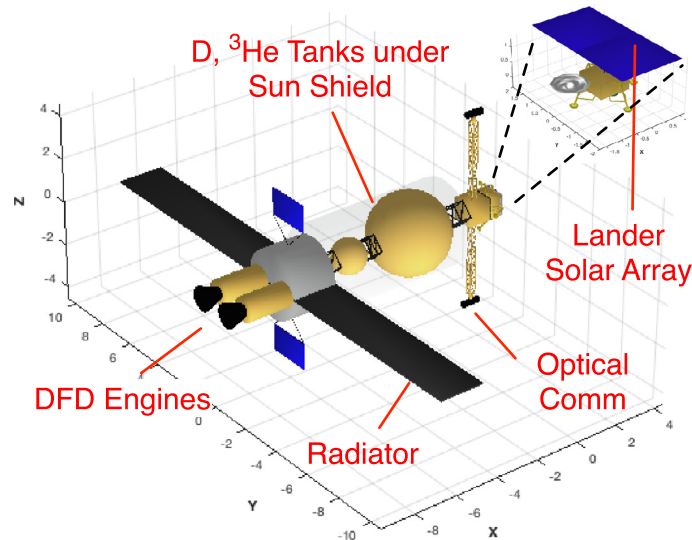


Figure 2.5 Pluto orbiter. DFD stands for Direct Fusion Drive.

The United States is funding three commercial companies to build human lunar landers. NASA is also funding SpaceX to build commercial lunar landers.

A large area of interest is in Low-Earth Orbit constellations. These involve hundreds, if not thousands, of satellites. For communications, the satellites must connect to each other, to ground stations, and user terminals on the ground. Example 2.2 shows two types of constellations. SpaceX is flying an enormous constellation for the Internet called Starlink. Amazon plans a similar constellation.

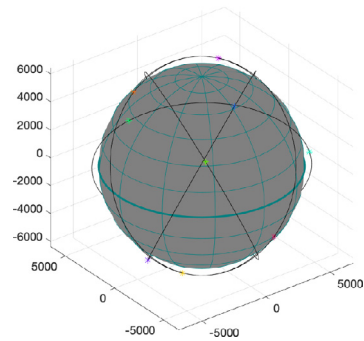
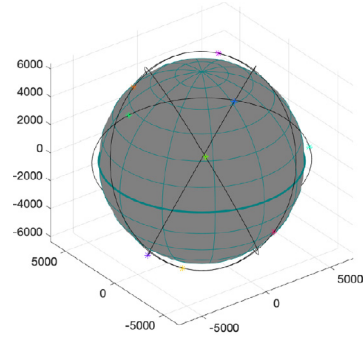
Space exploration has expanded beyond its beginnings as US/Soviet competition. It is now a truly international effort. China landed a lander with a rover on the far side of the moon and has launched communications, navigation, and scientific satellites into Earth orbit. The United Arab Emirates sent a mission to orbit Mars. India has also launched Mars missions and built a full range of satellites for every application. The European Space Agency (ESA) has built many advanced satellites, including lunar orbiters, communication satellites, and navigation satellites. The Swedish Space Corporation flew an ambitious rendezvous robots experiment and the first electric-propelled lunar mission. The ESA is building the service module for the Orion spacecraft for the Artemis lunar missions.

From a controls perspective, spacecraft have gone from passive control to analog control to digital control. Autonomy has always been a spacecraft requirement but even today many spacecraft require extensive interaction with the ground for operation. As the missions get more complex there will need to be more sophisticated onboard systems

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1 %% Walker Constellation
2
3 WalkerConstellation

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Example 2.2: Walker and polar-star constellations.

for control, fault detection, and mission planning. This book will help you build your background to address future challenges.

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